The visual beauty of 3DMark soothes the boredom of lengthy benchmarks

Going haywire isn't the answer; full-throttling resolution and other settings only heighten the risk of the testing process getting interrupted by a crash.

Keeping it conservative is vital not just for stability, but also for the scores to scale better in comparison. It is also an excellent method for pointing out performance gains from tweaks and hardware upgrades.

System Prerequisites

In order to run 3DMark2001 SE, your PC must meet some of today's prevalent standards. It's nothing that a two-year-old computer cannot match, aside from the graphics card perhaps. That needs to be current, of course. Here's the minimum you need: 500 MHz processor, 128 MB RAM, 100 MB of free hard disk space, DirectX 8.1 coupled with a 3D accelerator fitted with 32 MB of local memory

That last specification needs a little clarity. Since 3DMark 2001 investigates the performance of pixel and vertex shaders, it requires DirectX 8.1 to be installed. Also, it only makes sense to run it on a card that supports such graphical facets. And the eligible contestants are few; only the Geforce3, Radeon 8500 and later cards will do.

If you own anything that's at least one generator earlier, you may wish to use something else such as *Quake III* to benchmark your graphics card.

Making sense of the numbers

You've just seen how to benchmark and examine the performance of various systems that work in your PC. Furthermore, we've presented some comparative basis for you to work with. If you find that your numbers don't match up to the expected level, there could be some problems that needed fixing.

Starting with the video subsystem, there could be various minor issues that you could fix before embarking on an upgrade. A simple driver upgrade could fetch you a decent 10 per cent improvement. At the same time these driver upgrades will never show any significant improvements if the system is already heavily bottlenecked. A slower CPU and memory subsystem can be the worst companion to a graphics card.

Any amount of tweaking under those circumstances will achieve no improvement in performance unless you upgrade the core CPU and memory subsystem. A system running on a FSB of 66 MHz simply cannot keep up with modern applications and games. Try changing the drive to a faster 7200-rpm disk but you will find no improvement in loading times. These benchmarks will show you how the core subsystem of your PC is affecting various other subsystems and vice versa.

A thorough analysis of the scores can lead you to the right bottleneck, showing you exactly which subsystem is falling behind others. As an example, try running a GeForce3 card in a 500 MHz system and observe all game benchmarks. You will find all scores hitting a wall, no major improvement despite lowering resolutions, reducing quality and so on.

But, if everything is in order inside your PC and you still find yourself dissatisfied with the results, then there is another way we can help you. For performance boosts and tweaks on every major system in your PC, turn to our Tips & Tricks section. It's perfectly themed for those of you who want more speed and won't settle for anything less.

PRASHANT MASRANI